

CH365 Chemical Engineering Thermodynamics

Lesson 14

Generalized Correlations for Gases and Liquids

Generalized Correlations for Gases Slide 2

Lee-Kesler Method

Byung Ik Lee and Michael Kesler,
AIChE Journal, 1975, 21(3), 511-527

Pitzer Correlation for Z (L13)

(Eq. 3.53)

$$Z = Z^0 + \omega Z^1$$

simple fluids

deviation from
simple fluids:

$$\omega Z^1 = Z - Z^0$$

$$Z^0 = 1 + B^0 \frac{P_r}{T_r}$$

$$Z^1 = B^1 \cdot \frac{P_r}{T_r}$$

(Eq. 3.60)

Lesson 13, Slide 5
formulas for B^0 and B^1
eqns. 3.61 and 3.62

$$Z = 1 + \hat{B} \frac{P_r}{T_r} \text{ (eq. 3.57)}$$

$$\hat{B} = B^0 + \omega B^1 \text{ (eq. 3.59)}$$

$$\begin{aligned} Z &= 1 + (B^0 + \omega B^1) \frac{P_r}{T_r} \\ &= 1 + B^0 \frac{P_r}{T_r} + \omega B^1 \frac{P_r}{T_r} \\ &= Z^0 + \omega Z^1 \text{ (eq. 3.53, implied)} \end{aligned}$$

Lee-Kesler Modification

$Z^{(r)}$: calculated for n-octane

$$Z = Z^0 + \frac{\omega}{\omega^{(r)}} (Z^{(r)} - Z^0) \quad \text{where} \quad Z^1 = \frac{(Z^{(r)} - Z^0)}{\omega^{(r)}}$$

Lee and Kesler used a modified Benedict-Webb-Rubin EOS:

$$Z = 1 + \frac{B}{V_r} + \frac{C}{V_r^2} + \frac{D}{V_r^5} + \frac{c_4}{T_r^3 V_r^2} \left(\beta + \frac{\gamma}{V_r^2} \right) \exp \left(-\frac{\gamma}{V_r^2} \right)$$

B, C, D are functions of T_r
(published in the paper)

β, γ, c_4 , etc. are constants

Z calculated twice:

Z^0 : calculated for simple fluids

$Z^{(r)}$: calculated for n-octane

Tables: Appendix – Tables D.1-D.4, pp. 676-692

Example: Find Z for n-octane at $P_r=0.4$, $T_r=0.9$

Copyright © McGraw-Hill Education. Permission required for reproduction or display.

Table D.1: Values of Z^0 Page 677

$P_r =$	0.0100	0.0500	0.1000	0.2000	0.4000	0.6000	0.8000	1.0000
T_r								
0.30	0.0029	0.0145	0.0290	0.0579	0.1158	0.1737	0.2315	0.2892
0.35	0.0026	0.0130	0.0261	0.0522	0.1043	0.1564	0.2084	0.2604
0.40	0.0024	0.0119	0.0239	0.0477	0.0953	0.1429	0.1904	0.2379
0.45	0.0022	0.0110	0.0221	0.0442	0.0882	0.1322	0.1762	0.2200
0.50	0.0021	0.0103	0.0207	0.0413	0.0825	0.1236	0.1647	0.2056
0.55	0.9804	0.0098	0.0195	0.0390	0.0778	0.1166	0.1553	0.1939
0.60	0.9849	0.0093	0.0186	0.0371	0.0741	0.1109	0.1476	0.1842
0.65	0.9881	0.9377	0.0178	0.0356	0.0710	0.1063	0.1415	0.1765
0.70	0.9904	0.9504	0.8958	0.0344	0.0687	0.1027	0.1366	0.1703
0.75	0.9922	0.9598	0.9165	0.0336	0.0670	0.1001	0.1330	0.1656
0.80	0.9935	0.9669	0.9319	0.8539	0.0661	0.0985	0.1307	0.1626
0.85	0.9946	0.9725	0.9436	0.8810	0.0661	0.0983	0.1301	0.1614
0.90	0.9954	0.9768	0.9528	0.9015	0.7800	0.1006	0.1321	0.1630
0.93	0.9959	0.9790	0.9573	0.9115	0.8059	0.6635	0.1359	0.1664
0.95	0.9961	0.9803	0.9600	0.9174	0.8206	0.6967	0.1410	0.1705
0.97	0.9963	0.9815	0.9625	0.9227	0.8338	0.7240	0.5580	0.1779
0.98	0.9965	0.9821	0.9637	0.9253	0.8398	0.7360	0.5887	0.1844
0.99								
1.00								
1.01								
1.02								
1.05								
1.10								
1.15								
1.20								
1.30								
1.40								
1.50								
1.60								
1.70								
1.80								
1.90								
2.00								
2.20								
2.40								
2.60								
2.80								
3.00								
3.50								
4.00								

$Z^0=0.780$

$Z^1=-0.1118$

$\omega=0.400$

$$Z = Z^0 + \omega Z^1$$
$$= 0.780 + (0.400) \cdot (-0.1118)$$
$$= .7353$$

Copyright © McGraw-Hill Education. Permission required for reproduction or display.

Table D.2: Values of Z^1 Page 678

$P_r =$	0.0100	0.0500	0.1000	0.2000	0.4000	0.6000	0.8000	1.0000
T_r								
0.30	-0.0008	-0.0040	-0.0081	-0.0161	-0.0323	-0.0484	-0.0645	-0.0806
0.35	-0.0009	-0.0046	-0.0093	-0.0185	-0.0370	-0.0554	-0.0738	-0.0921
0.40	-0.0010	-0.0048	-0.0095	-0.0190	-0.0380	-0.0570	-0.0758	-0.0946
0.45	-0.0009	-0.0047	-0.0094	-0.0187	-0.0374	-0.0560	-0.0745	-0.0929
0.50	-0.0009	-0.0045	-0.0090	-0.0181	-0.0360	-0.0539	-0.0716	-0.0893
0.55	-0.0314	-0.0043	-0.0086	-0.0172	-0.0343	-0.0513	-0.0682	-0.0849
0.60	-0.0205	-0.0041	-0.0082	-0.0164	-0.0326	-0.0487	-0.0646	-0.0803
0.65	-0.0137	-0.0072	-0.0078	-0.0156	-0.0309	-0.0461	-0.0611	-0.0759
0.70	-0.0093	-0.0507	-0.1161	-0.0148	-0.0294	-0.0438	-0.0579	-0.0718
0.75	-0.0064	-0.0339	-0.0744	-0.0143	-0.0282	-0.0417	-0.0550	-0.0681
0.80	-0.0044	-0.0228	-0.0487	-0.1160	-0.0272	-0.0401	-0.0526	-0.0648
0.85	-0.0029	-0.0152	-0.0319	-0.0715	-0.0268	-0.0391	-0.0509	-0.0622
0.90	-0.0019	-0.0099	-0.0205	-0.0442	-0.1118	-0.0396	-0.0503	-0.0604
0.93	-0.0015	-0.0075	-0.0154	-0.0326	-0.0763	-0.1662	-0.0514	-0.0602
0.95	-0.0012	-0.0062	-0.0126	-0.0262	-0.0589	-0.1110	-0.0540	-0.0607
0.97	-0.0010	-0.0050	-0.0101	-0.0208	-0.0450	-0.0770	-0.1647	-0.0623
0.98	-0.0009	-0.0044	-0.0090	-0.0184	-0.0390	-0.0641	-0.1100	-0.0641
0.99	-0.0008	-0.0039	-0.0079	-0.0161	-0.0335	-0.0531	-0.0796	-0.0680
1.00	-0.0007	-0.0034	-0.0069	-0.0140	-0.0285	-0.0435	-0.0588	-0.0879
1.01	-0.0006	-0.0030	-0.0060	-0.0120	-0.0240	-0.0351	-0.0429	-0.0223
1.02	-0.0005	-0.0026	-0.0051	-0.0102	-0.0198	-0.0277	-0.0303	-0.0062
1.05	-0.0003	-0.0015	-0.0029	-0.0054	-0.0092	-0.0097	-0.0032	0.0220
1.10	0.0000	0.0000	0.0001	0.0007	0.0038	0.0106	0.0236	0.0476
1.15	0.0002	0.0011	0.0023	0.0052	0.0127	0.0237	0.0396	0.0625
1.20	0.0004	0.0019	0.0039	0.0084	0.0190	0.0326	0.0499	0.0719
1.30	0.0006	0.0030	0.0061	0.0125	0.0267	0.0429	0.0612	0.0819
1.40	0.0007	0.0036	0.0072	0.0147	0.0306	0.0477	0.0661	0.0857
1.50	0.0008	0.0039	0.0078	0.0158	0.0323	0.0497	0.0677	0.0864
1.60	0.0008	0.0040	0.0080	0.0162	0.0330	0.0501	0.0677	0.0855
1.70	0.0008	0.0040	0.0081	0.0163	0.0329	0.0497	0.0667	0.0838
1.80	0.0008	0.0040	0.0081	0.0162	0.0325	0.0488	0.0652	0.0814
1.90	0.0008	0.0040	0.0079	0.0159	0.0318	0.0477	0.0635	0.0792
2.00	0.0008	0.0039	0.0078	0.0155	0.0310	0.0464	0.0617	0.0767
2.20	0.0007	0.0037	0.0074	0.0147	0.0293	0.0437	0.0579	0.0719
2.40	0.0007	0.0035	0.0070	0.0139	0.0276	0.0411	0.0544	0.0675
2.60	0.0007	0.0033	0.0066	0.0131	0.0260	0.0387	0.0512	0.0634
2.80	0.0006	0.0031	0.0062	0.0124	0.0245	0.0365	0.0483	0.0598
3.00	0.0006	0.0029	0.0059	0.0117	0.0232	0.0345	0.0456	0.0565
3.50	0.0005	0.0026	0.0052	0.0103	0.0204	0.0303	0.0401	0.0497
4.00	0.0005	0.0023	0.0046	0.0091	0.0182	0.0270	0.0357	0.0443

Generalized Correlations for Liquids Slide 4

Rackett: $V^{\text{sat}} = V_C Z_C^{(1-T_r)^{2/7}}$ (Eq. 3.68)

$$Z^{\text{sat}} = \frac{P_r}{T_r} Z_C^{[1+(1-T_r)^{2/7}]} \quad (\text{Eq. 3.69})$$

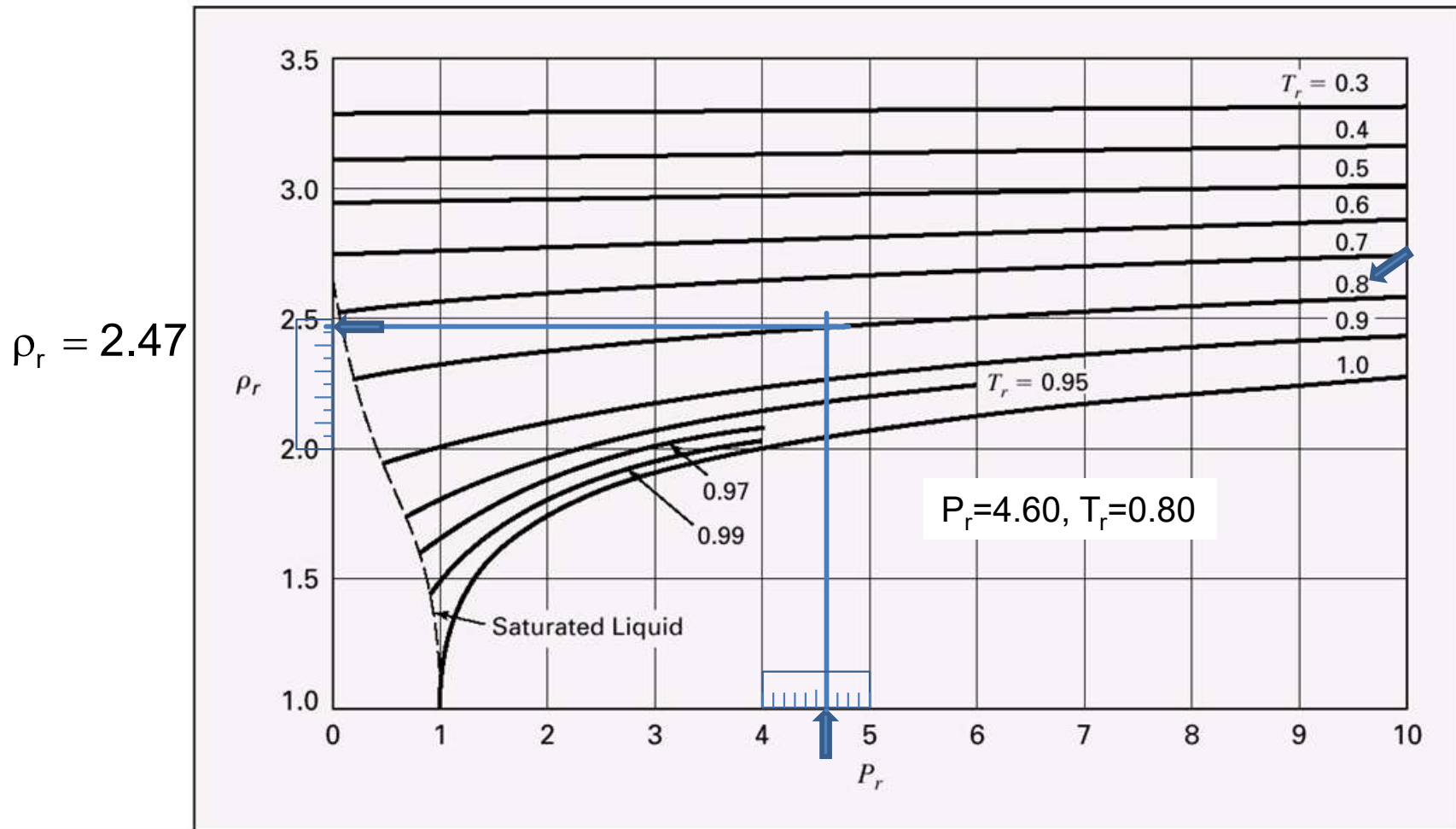


Figure 3.15: Generalized density correlation for liquids.

Lydersen, Greenkorn, and Hougen: $\rho_r \equiv \frac{\rho}{\rho_C} = \frac{V_C}{V}$ (Eq. 3.70)

Example 3.14

For ammonia at 310 K, estimate the molar volume density of (a) the saturated liquid and (b) the liquid at 100 bar.

experimental:

$$V^{\text{sat}} = 29.14 \frac{\text{cm}^3}{\text{mol}}$$

Part (a) Use Table B.1, Appendix B, pp. 663-665:

$$(B.1, 665) \quad T_C = 405.7 \text{ K} \Rightarrow T_r = \frac{310}{405.7} = 0.7641$$

$$(B.1, 665) \quad V_C = 72.5 \frac{\text{cm}^3}{\text{mol}}$$

$$(B.1, 665) \quad Z_C = 0.242$$

(Eq. 3.68, Rackett)

$$\left. \begin{array}{l} (B.1, 665) \quad V_C = 72.5 \frac{\text{cm}^3}{\text{mol}} \\ (B.1, 665) \quad Z_C = 0.242 \end{array} \right\} \Rightarrow V^{\text{sat}} = V_C Z_C^{(1-T_r)^{2/7}} = 72.5 \cdot 0.242^{(1-0.7641)^{2/7}} = 28.35 \frac{\text{cm}^3}{\text{mol}}$$

ANS

Part (b)

$$(B.1, 665) \quad P_C = 112.8 \text{ bar} \Rightarrow P_r = \frac{100}{112.8} = 0.8865$$

$$\Rightarrow \begin{array}{l} T_r = 0.7641 \\ P_r = 0.8865 \end{array}$$

➡ Now use Fig. 3.15 for ρ_r then eq. 3.70 for molar volume

(Liquids)

Example 3.14 , part b, continued

Rackett: $V^{\text{sat}} = V_C Z_C^{(1-T_r)^{2/7}}$ (Eq. 3.68) $Z^{\text{sat}} = \frac{P_r}{T_r} Z_C^{[1+(1-T_r)^{2/7}]}$ (Eq. 3.69)

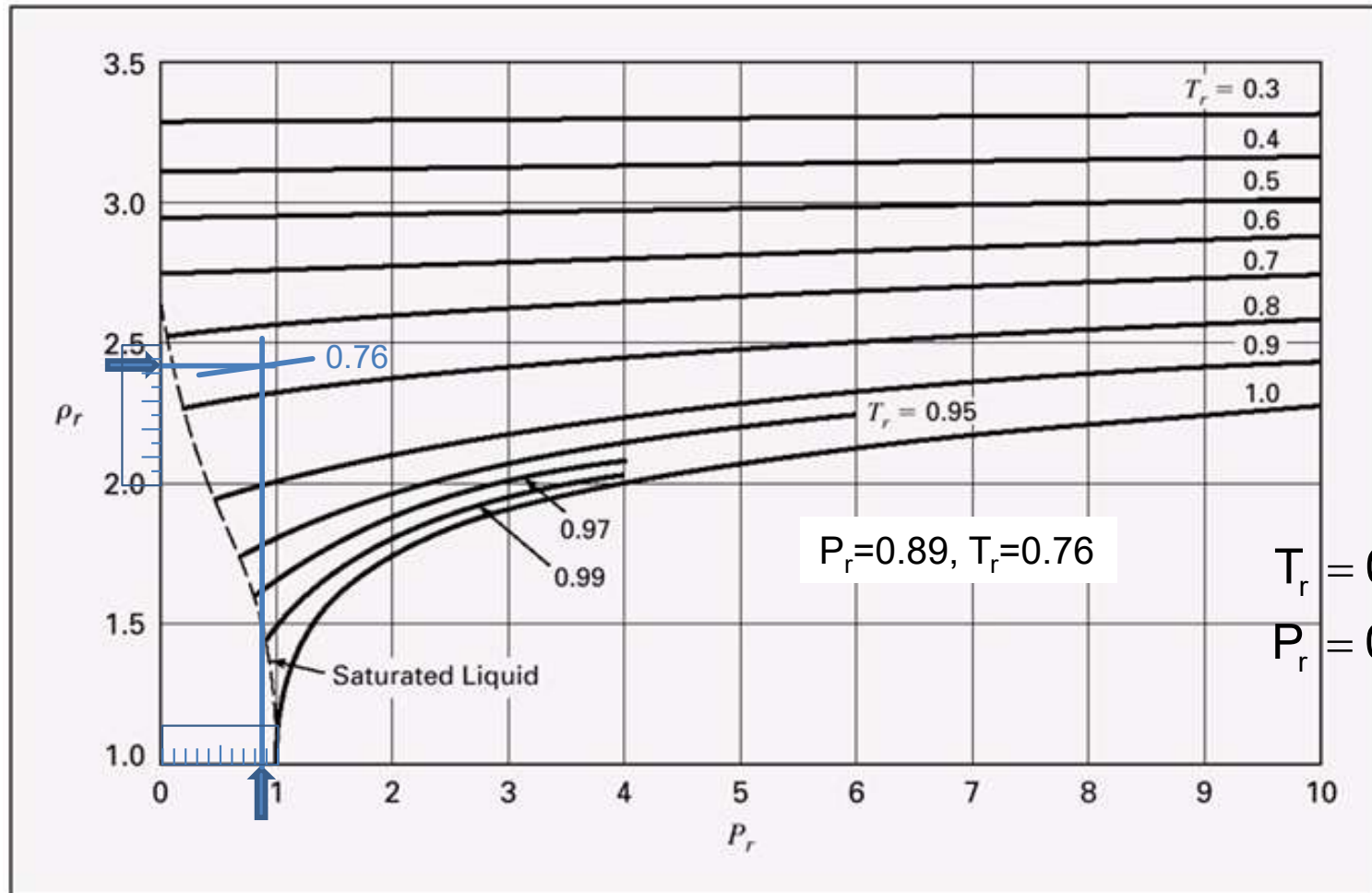


Figure 3.15. Generalized density correlation for liquids.

Lydersen, Greenkorn, and Hougen:

$$\rho_r \equiv \frac{\rho}{\rho_C} = \frac{V_C}{V} \quad (\text{Eq. 3.70})$$

$$\rho_r = 2.43$$

solution deviates
from book;
authors read
reduced density
as 2.38 on p. 117

$$T_r = 0.7641$$

$$P_r = 0.8865$$

Example 3.14, continued

For ammonia at 310 K, estimate the molar volume density of (a) the saturated liquid and (b) the liquid at 100 bar.

experimental:

$$V^{\text{sat}} = 29.14 \frac{\text{cm}^3}{\text{mol}}$$

Part (a) Use Table B.1, Appendix B, pp. 663-665:

$$(B.1, 665) \quad T_C = 405.7 \text{ K} \Rightarrow T_r = \frac{310}{405.7} = 0.7641$$

$$(B.1, 665) \quad V_C = 72.5 \frac{\text{cm}^3}{\text{mol}}$$

$$(B.1, 665) \quad Z_C = 0.242$$

(Eq. 3.68, Rackett)

$$\left. \begin{array}{l} (B.1, 665) \quad V_C = 72.5 \frac{\text{cm}^3}{\text{mol}} \\ (B.1, 665) \quad Z_C = 0.242 \end{array} \right\} \Rightarrow V^{\text{sat}} = V_C Z_C^{(1-T_r)^{2/7}} = 72.5 \cdot 0.242^{(1-0.7641)^{2/7}} = 28.35 \frac{\text{cm}^3}{\text{mol}}$$

ANS

Part (b)

$$P_C = 112.8 \text{ bar} \Rightarrow P_r = \frac{100}{112.8} = 0.8865 \Rightarrow \rho_r = 2.43 \quad \text{Use Fig. 3.15}$$

$$\rho_r \equiv \frac{\rho}{\rho_C} = \frac{V_C}{V} \Rightarrow V = \frac{V_C}{\rho_r} = \frac{72.5}{2.43} = 29.84 \frac{\text{cm}^3}{\text{mol}}$$

(Eq. 3.70)

$\rho_r = 2.43$ from slide 6, Figure 3.15

ANS

Questions